

Time mapper

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Time mapper

The Time mapper is an animated bar chart that shows the evolution of the frequency of a chosen variable across time. In input, the algorithm requires a csv file that contains either:

1. a field with header 'Document' that has dates imbedded in the filenames (e.g, New York Time_10-2-1941);
2. a field with header 'Date'.

Whenever a Date field is present it takes precedence over the Document field, even if the filename embeds a date.

1	Word	NER	tokenBegin	tokenEnd	Sentence	Sentence	Document ID	Document
2	first	ORDINAL	3	4	1	WHEN it w	1	C:\Users\
3	America	COUNTRY	15	16	1	WHEN it w	1	C:\Users\
4	Providence	CITY	29	30	2	Relying, ho	1	C:\Users\
5	first	ORDINAL	39	40	2	Relying, ho	1	C:\Users\
6	present	DATE	55	56	2	Relying, ho	1	C:\Users\
7	Revolution	IDEOLOGY	9	10	3	The zeal ar	1	C:\Users\
8	war	CAUSE_OF	10	11	3	The zeal ar	1	C:\Users\
9	Confedera	MISC	1	2	4	The Confec	1	C:\Users\
10	Batavian	LOCATION	16	17	4	The Confec	1	C:\Users\
11	courier	TITLE	17	18	5	But reflect	1	C:\Users\
12	day	DURATION	31	32	5	But reflect	1	C:\Users\
13	Congress	ORGANIZA	43	44	5	But reflect	1	C:\Users\

Time mapper GUI and input

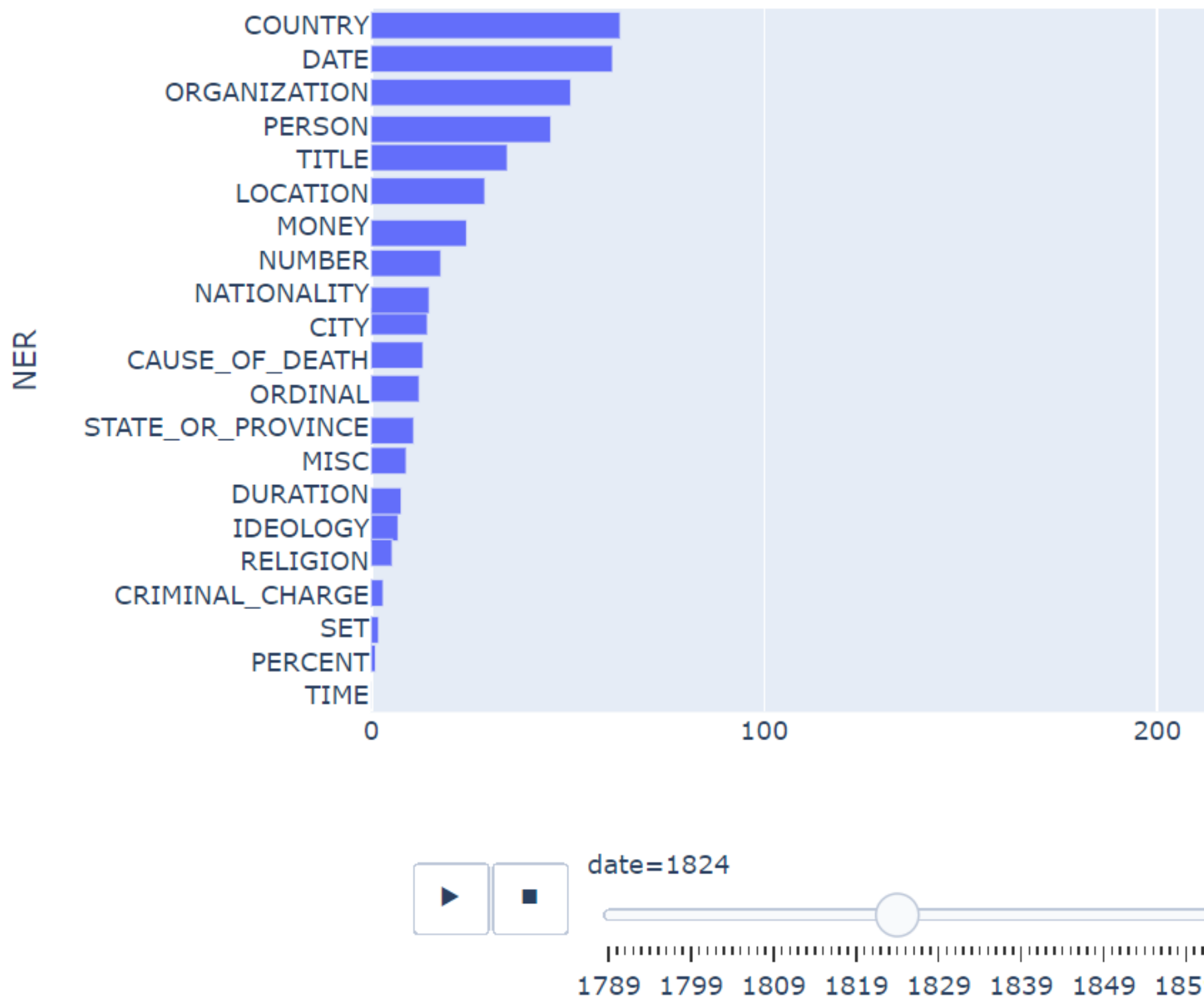
? HELP	csv file field for visualization (Y-axis)	NER
? HELP	Visualization options	Time mapper
? HELP	Excel/Plotly	X-axis
? HELP	Boxplot	Data
? HELP	Bar charts	+ Reset
? HELP	Time mapper	Date format mm-dd-yyyy

The GUI for the time mapper requires the selection of:

1. the variable that you want to visualize using the dropdown menu for csv file field for visualization (Y-axis); **the selected variable should be categorical, or at least have only a few values not to clutter the output;**
2. the appropriate date format contained in the date embedded in the filename under Document or in the Date field, using the dropdown menu for Date format;
3. the desired timeline (daily, monthly, yearly);
4. whether the output displayed by the visualization algorithm should be cumulative.

Time mapper output

The following example is **yearly cumulative data of the NER variable from the CoNLL table** using the POTUS inaugural addresses and State of the Union speeches.



Once you obtain your output, the triangular button will automatically run through all the dates, the square button will stop the animation, and the cursor underneath the graph will allow you to manually drag the date circle left-right and start visualizing data from the selected date onward. You can also run your mouse over the bars to see the current values of the frequencies.

The values displayed may change as the timeline moves forward, thus DATE is on top, i.e., more frequent than COUNTRY, at some point in time.

DATE
ORGANIZATION
COUNTRY